Samuel Pereira

Music Theory · Computational Musicology

Matosinhos, Portugal · +351 916199033 · sml.p@outlook.pt · Website · ORCID

I'm a researcher in music theory and computational musicology developing the Fourier Qualia Space, a novel framework applying discrete Fourier transforms to model harmonic-quality motion and ambiguity across four centuries of Western art music, with a special focus on post-tonal repertoires. My work combines quantitative analysis with music-theoretical enquiry, bridging computational methods and humanistic interpretation to understand how harmonic qualities convey hierarchical and syntactic organisation.

ACADEMIC AND RESEARCH POSITIONS

Assistant Professor

2026–

University of Coimbra

Teaching across two master's programmes in two faculties:

- *Computer Music and Sound Design*. Department of Informatics Engineering (DEI), Faculty of Sciences and Technology;
- *Art Studies*. Faculty of Arts and Humanities.

Researcher

2023–2024

Marie Skłodowska-Curie Actions (MSCA) Project "EA-DIGIFOLK"

- Project title: *An European and Ibero-American approach for the digital collection, analysis and dissemination of folk music*.
- Provided expertise on computational approaches to musical similarity, with a special focus on folk music similarity perception. Completed two secondments at the University of Salamanca, Spain, contributing insights on how musicians conceptualise and perceive harmonic, melodic, rhythmic, textural, and formal relationships.

External Student

2021–

INESC TEC, Porto, Portugal

- Conducting doctoral research on symbolic harmonic analysis using discrete Fourier transforms. Developed computational tools for large-scale corpus analysis and harmonic qualities visualisation. Contributed to research outputs including peer-reviewed publications on musical syntax and open-source software.

EDUCATION

DEGREES

PhD in Digital Media — 2021–2026

University of Porto, Faculty of Engineering (FEUP)

- Thesis title: *The Fourier Qualia Space: Interaction, Ambiguity, and Hierarchy in Music Harmony*
- Supervisors: Prof. Gilberto Bernardes, Prof. José Oliveira Martins
- Public defence: June 15th 2026
- Approved by unanimity
- Examination Committee: Dr. António Coelho, Dr. Martin Rohrmeier, Dr. Matt Chiu, Dr. Isabel Pires, Dr. Gilberto Bernardes, and Dr. António Sá Pinto

Master's in Multimedia — 2018–2020

University of Porto, Faculty of Engineering (FEUP)

- Dissertation title: *The Atonal Harmonic Syntax in Fourier Space: Towards a Spherical Representation of Phases*
- Supervisor: Prof. Gilberto Bernardes
- Final grade: 18/20

Bachelor's in Musical Composition — 2014–2017

Escola Superior de Música e Artes do Espectáculo (ESMAE)

- Final grade: 17/20

CERTIFICATIONS

Certificado de Competências Pedagógicas (CCP) — 2021

Escola de Negócios e Administração (ENA)

- Professional teaching certification for vocational training
- Final grade: 5/5

GRANTS AND AWARDS

Doctoral Grant

2021–2025

Fundação para a Ciência e a Tecnologia (FCT)

- Project reference: 2021.05059.BD
- Title: *Cartografias Harmónicas e a Conceptualização do Movimento Musical: Uma Sintaxe para a Música Atonal no Espaço de Fourier*
- Four-year funding for doctoral research in computational musicology, investigating whether post-tonal music possesses a coherent syntax in the unfolding of harmonic qualities (e.g., chromaticity, octatonicity, and diatonicity) over time.

PUBLICATIONS

JOURNAL ARTICLES

Pereira, Samuel, Gilberto Bernardes, and José Oliveira Martins. *"Statistically Modelling Qualia Syntax Through Fourier Space: A Three-role Probabilistic Framework Where Ambiguity Becomes Structure"* — Forthcoming

Musicae Scientiae

Pereira, Samuel, Gilberto Bernardes, and José Oliveira Martins. *"Qualia Motion in Fourier Space: Formalising Linear, Nondirected and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1"* — 2025

Music Theory Spectrum 48 (1).

Bernardes, Gilberto, Nádia Carvalho, and Samuel Pereira. *"FluidHarmony: Defining an Equal-Tempered and Hierarchical Harmonic Lexicon in the Fourier Space"* — 2022

Journal of New Music Research 51 (2–3): 142–61.

BOOK CHAPTERS

Pereira, Samuel, Giovanni Affatato, Gilberto Bernardes, and Fabian C. Moss. *"Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity"* — 2024

In **Mathematics and Computation in Music**, ed. Thomas Noll et al. Springer.

SOFTWARE AND DATA

Pereira, Samuel. *sml-pereira/phd-fqs: v1.0.1 — Repository Release* 2026

Zenodo.

Pereira, Samuel. *sml-pereira/qualia-syntax: v1.0.0 — Repository Release* 2026

Zenodo.

CONFERENCE PAPERS IN PROCEEDINGS

Pereira, Samuel and Gilberto Bernardes. *"The Fourier Phase Sphere: A Method for Computer-Assisted Atonal Composition"* 2022

5th International Conference on Music and Mathematics (MusMat 2020), Perspectives and Applications of Mathematics in Post-tonal Theories («Homage to Jamary Oliveira»), online edition, pp. 40–54

CONFERENCE PRESENTATIONS

"The Fourier Qualia Space: Interaction, Ambiguity, and Hierarchy in Music Harmony" Sept. 2026

EuroMAC 11: 11th European Music Analysis Conference, Escola Superior de Música de Catalunya (ESMUC), Barcelona

"Historical Evolution of Harmonic Qualities in Music: Unravelling Fourier's Tonality, Entropy, and Ambiguity from the 16th to the 20th Centuries" 2024

Digital Technologies Applied to Music Research Conference: Methodologies, Projects and Challenges, Faculty of Social Sciences and Humanities – NOVA University, Lisbon

"Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity" 2024

9th International Congress on Mathematics and Computation in Music (MCM2024), Faculty of Arts and Humanities of the University of Coimbra, Coimbra

"A Propósito da Representação do Movimento Musical: o Tonnetz e os Espaços de Fourier" 2022

XI ENIM – Encontro Nacional de Investigação em Música, University of Aveiro, Aveiro.

"The Atonal Harmonic Syntax in Fourier's Space" 2021

X ENIM – Encontro Nacional de Investigação em Música, Faculty of Arts and Humanities of the University of Coimbra, Coimbra

"Harmonic Cartographies and the Conceptualization of Musical Movement: A Syntax for Atonal Music in Fourier Space" 2021

Reboot Fest 2021 – New Media and Digital Art Festival, Porto

"The Fourier Phase Sphere: A Method for Computer-Assisted Atonal Composition" 2020

5th International Conference on Music and Mathematics (MusMat 2020), online

EDITORIAL AND PEER REVIEW

Reviews Editor *(Editor de Recensões)* 2026–

Revista Portuguesa de Musicologia · ISSN 2183-8410

· Published in partnership by SPIM (Sociedade Portuguesa de Investigação em Música), INET-md (Instituto de Etnomusicologia — Centro de Estudos em Música e Dança), and CESEM (Centre for Music Studies).

TECHNICAL SKILLS

Programming	Python (`NumPy`, `SciPy`, `Pandas`, `Matplotlib`, `scikit-learn`, `Seaborn`, `tqdm`, `music21`), LaTeX, Shell scripting
Statistics	Discrete Fourier transform analysis, PCA, t-SNE, hierarchical clustering, Markov chain analysis, n-gram modelling, entropy metrics, Zipf's law
Data	Large-scale corpus management, automated pipeline development, feature extraction
Visualisation	Publication-quality figures, heatmaps, dendrograms, wavescapes, interactive plots
Tools	Jupyter Notebook, Git, VS Code, Claude AI, Julius AI, Tableau, Jamovi, RAWGraphs, Overleaf, Zenodo, Office Suite, Sibelius, MuseScore, Pure Data, Max/MSP, Reaper

LANGUAGES

Portuguese *native* · English *fluent* · Spanish *reading proficiency*

OTHER ACTIVITIES

European Youth Parliament

2005–2006

- National session delegate, Covilhã, Portugal.
- International session delegate, Paris, France.